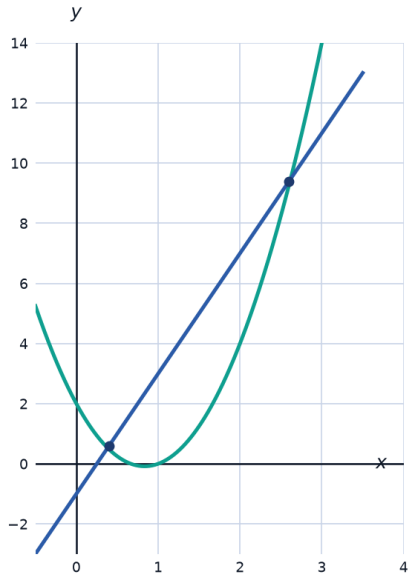
Calculator & Test Strategy

Section 1: Graphical Systems & Intersections

1 The graphs of $y = 3x^2 - 5x + 2$ and $y = 4x - 1$ intersect at two points. What is the sum of the x-coordinates of the intersection points?



a. 2

b. 3

c. 4

d. 5

2 A graphing calculator displays the roots of $y = x^3 - 4x^2 - x + 4$. How many distinct real zeros does the function possess?

a. 1

b. 2

c. 3

d. 4

3 For what value of c does the horizontal line $y = c$ intersect the parabola $y = -(x - 3)^2 + 8$ at exactly one point?

a. 3

b. 5

c. 8

d. 11

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Section 2: Slider Tactics & Variable Parameters

4 If the equation $2x + a = 3(x - 2) + 4$ is true for $x = 5$, what is the value of a?

a. 3

b. 5

c. 7

d. 9

5 In the equation $\frac{4x+k}{2} = 2x + 7$, what value of k makes the equation true for all values of x?

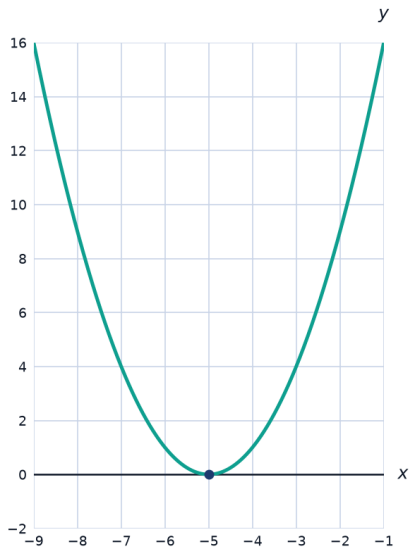
a. 7

b. 14

c. 21

d. 28

6 For which value of b does the quadratic equation $x^2 + bx + 25 = 0$ have only one distinct real solution?



a. 5

b. 10

c. 15

d. 20

Section 3: Table Verification & Sequence Checks

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7 The table of values shows that $f(1) = 5, f(2) = 11, f(3) = 17$. If f is a linear function, what is the value of $f(10)$?

a. 53

b. 59

c. 65

d. 71

8 A function satisfies $g(x) = 2^{x+1}$. What is the value of $g(4) - g(2)$?

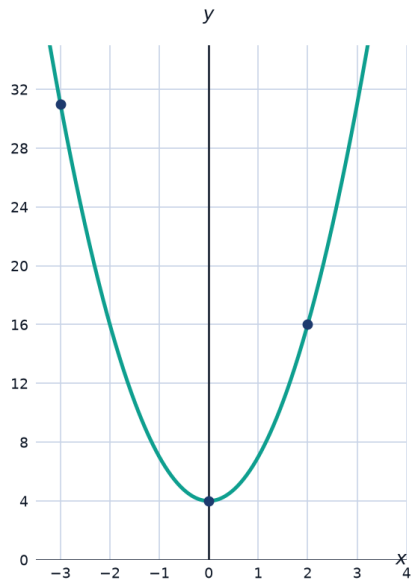
a. 16

b. 24

c. 32

d. 48

9 If $h(x) = ax^2 + b$, with $h(0) = 4$ and $h(2) = 16$, what is the value of $h(-3)$?



a. 25

b. 31

c. 36

d. 40

Section 4: Strategic Approximations & Back-Solving

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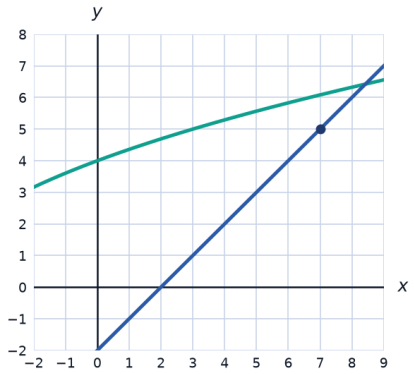
10 What is the positive solution to $\sqrt{3x+16} = x - 2$?

a. 4

b. 5

c. 6

d. 7



11 If $x > 0$ and $x^4 - 13x^2 + 36 = 0$, what is the sum of the positive real solutions?

a. 3

b. 5

c. 6

d. 8

12 If $\frac{x}{x-3} - \frac{2}{x+1} = 1$, what is the value of x?

a. 1

b. 2

c. 3

d. 5

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